

New York City TLC Project - Preliminary Data Analysis

By **Automatidata**

Overview

At this stage of this project, a preliminary analysis of the TLC data is performed to understand the key data variable and verify the suitability of information for generating insights and predictive model.

Key Insights

- This dataset includes variables that seems relevant in building predictive model.
- Unusual value (-negative amount) is noticed for total_amount for some records.
- The passenger_count does not seems correlated with trip_distance.

trip_distance		total_amount		tolls_amount	
9280	33.96	8476	1200.29	5271	19.10
13861	33.92	20312	450.30	16705	18.28
6064	32.72	13861	258.21	4885	18.26
10291	31.95	12511	233.74	13359	18.00
29	30.83	15474	211.80	18888	18.00
18130	30.50	...	...	11560	17.50
5792	30.33	11204	-5.30	7627	17.28
15350	28.23	14714	-5.30	17959	16.62
10302	28.20	17602	-5.80	316	16.50
2592	27.97	20698	-5.80	17046	16.50
20612	27.88	12944	-120.30	6064	16.26
1908	27.34	trip_distance		16379	16.26
20545	27.20	passenger		21	16.26
4138	26.86	0	2.803704	15421	16.20
15169	26.54	1	2.981663	17111	16.00
1496	26.39	2	3.322412	10875	16.00
7217	26.20	3	3.084283	7929	15.58
908	26.12	4	2.867341	5536	15.50
19483	26.12	5	3.007561	7746	15.50
4715	25.86	6	3.136319	2478	15.00

Details

- We have investigated a 0.02% sample data of TLC complete dataset.
- Explored dataset to find any unusual value.
- trip_distance, total_amount, and tolls_amount seems most useful to build model(s) for predicting trip_duration.
- Investigate the relationship between two chosen variables.
- Built up the basic understanding of TLC dataset for future exploratory data analysis, insights, visualizations, and models.
- We will perform EDA as next step of this project

Next Steps

1. Perform Exploratory Data Analysis on Complete Dataset.
2. Use Data Cleaning and Manipulation for predictive modeling.
3. Perform Data Wrangling on data variables like datetime that support predictive modeling.
4. We can further analyse variables correlation to find best variables for predictive model(s)